Software Requirements Specification (SRS)

Egypt Real-State Platform - 21 website

1 Introduction

The Egy Real-State Platform "21 website" is a bilingual (Arabic/English) digital real-estate ecosystem designed to support property browsing, project management, and customer engagement across multiple interfaces. The system provides both light and dark mode support and aims to streamline interactions between customers, managers, staff members, and real-estate developers.

The solution consists of:

- Public website for customers
- Mobile application (Flutter)
- Internal administrative dashboard (Manager and Staff)
- External Developer Portal
- Backend system integrated & tracking system and AI recommendation services

This document outlines the functional and non-functional requirements for 21 realstate website of the system.

2 Goal

The goal of this system is to deliver a secure, scalable, and high-quality real-estate platform that enables efficient listing, management, and exploration of real-estate units and projects. The platform integrates intelligent recommendation capabilities and provides seamless interaction across all user roles.

3 Objectives

- Develop a unified bilingual digital presence accessible via web and mobile.

- Deliver structured browsing capabilities, including filtering and comparison.
- Provide a standardized process for customer inquiries and lead capture.
- Enable managers and staff to manage platform content through a centralized dashboard.
- Provide developers with a portal to manage their own projects and units.
- Implement an AI-driven recommendation and user behavior tracking system..
- Maintain consistent UI/UX design with light/dark mode across all platforms.

4 Stakeholders

- **Customers**: Explore units, projects, developers, and submit inquiries.
- **Managers**: Full administrative access, system-wide control.
- **Staff**: Limited dashboard access for content management.
- **Real-Estate Developers**: Manage own listings via Developer Portal.

5 Functional Requirements

5.1 Customer Website and Mobile Application

5.1.1 Landing Page Requirements

- Display most popular units and projects.
- Display trending/popular areas.
- Display newest units and projects.
- Provide AI-powered recommendations based on user behavior.
- Display resale units.
- Display partner developers.
- Provide a global search bar with multi-parameter search.
- Provide quick-access communication options (WhatsApp, Messenger, Call).
- Include footer with standard informational sections.

5.1.2 Navigation Bar Requirements

- Access units listing.
- Access developers listing.

- Access projects listing.
- Access areas and countries.
- Enable users to compare units.
- Access "Who We Are" informational page.
- Provide Login/Register for unauthenticated users or Logout for authenticated users.

5.2 Projects Module

5.2.1 Projects Listing

- Display all projects in card format (title, location, price range, cover image).
- Provide filtering and search functionalities.

5.2.2 Project Details

- Provide image and video gallery.
- Display title, detailed description, and location map.
- Display available units under the project.
- Display pricing information and feature list.
- Provide rating functionality.
- Provide a customer inquiry form with auto-filled interest fields.

5.3 Developers Module

5.3.1 Developers Listing

- Display all developers with name, logo, and quick overview.
- Provide search and filtering options.

5.3.2 Developer Details

- Display detailed profile including name, description, and logo.
- Display all related projects.

5.4 Units Module

5.4.1 Units Listing

- Display units as cards (main image, title, price, developer, location, summary details).
- Provide advanced search and filtering.

5.4.2 Unit Details

- Display image gallery and descriptive content.
- Display unit features, technical specifications, and payment plans.
- Provide location map and optional video content.
- Provide customer inquiry form with pre-defined interest context.

5.5 Areas and Countries

- List all areas and countries with image and title.
- On selection: display related projects.
- Provide search and filtering.

6 Dashboard (Manager & Staff)

6.1 Dashboard Features

- Full CRUD management for units, projects, developers, areas, countries, features, and unit types.
- Access and manage submitted customer inquiries.
- View complete customer list.

6.2 User Roles

- **Manager**: Full system access including deletion and oversight functions.
- **Staff**: Restricted access limited to content creation and modification.

7 Developer Portal

- Allow developer login via secure authentication.
- Enable developers to create, update, and delete their own units and projects.
- Restrict access to developer-owned data only.
- Provide optional registration for new developer accounts.

8 CRM Integration

- Sync customer inquiries via API or webhook.
- Capture and forward user behavioral tracking for CRM workflows.

9 Non-Functional Requirements

9.1 Security Requirements

- Implement JWT-based authentication.
- Apply role-based access control (RBAC).
- Enforce API rate limiting to prevent abuse.
- Store passwords securely using hashing and salting.
- Require HTTPS for all service endpoints.
- Apply rigorous input validation for all API requests.
- Maintain audit logs for administrative actions.

9.2 Performance Requirements

- Maintain average API response time below 300ms.
- Use caching layers for frequently accessed content.
- Implement database indexing and optimized queries.
- Use pagination for all data-heavy endpoints.
- Run resource-intensive operations asynchronously (AI).
- Serve static media through CDN.

9.3 Quality Requirements

- Provide complete bilingual support (Arabic/English).
- Support both light and dark UI modes.
- Ensure consistent UI/UX across all platforms.
- System must scale to support a minimum of 5,000 daily active users.
- Maintain clean code practices: linting, documentation, automated testing.

10 External Interactions

- **Mobile Application (Flutter)**: Communicates via public API endpoints.
- **Customer Website**: Utilizes the same API layer.
- **Dashboard**: Accesses privileged admin endpoints.
- **Developer Portal**: Uses developer-restricted API routes.
- **CRM System**: Integrated through API/Webhook.
- **Messaging Platforms**: Integrated via deep links (WhatsApp, Messenger).
- **Map Service Provider**: Used for project and unit geolocation.
- **AI Engine**: Internal service consumes and processes user behavior data.